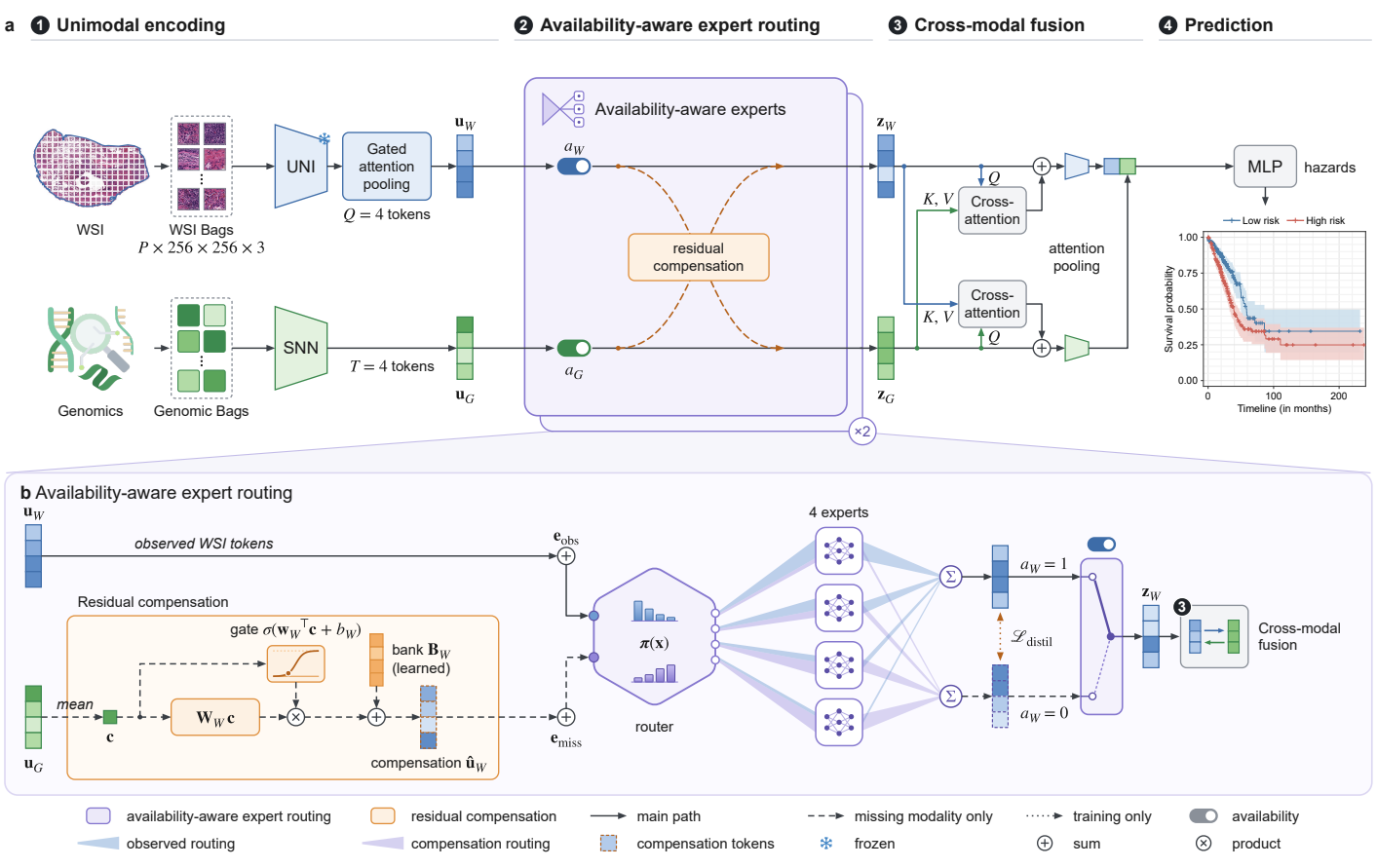

a
1 Unimodal encoding
2 Availability-aware expert routing
3 Cross-modal fusion
4 Prediction
WSI
WSI Bags
$P \times 256 \times 256 \times 3$
UNI
Gated attention pooling
$Q = 4$ tokens
$\mathbf{u}_W$
Genomics
Genomic Bags
SNN
$T = 4$ tokens
$\mathbf{u}_G$
Availability-aware experts
a_W
a_G
residual compensation
×2
$\mathbf{z}_W$
$\mathbf{z}_G$
Q
K, V
Cross-attention
Cross-attention
attention pooling
MLP
hazards
Low risk
High risk
Survival probability
Timeline (in months)
b Availability-aware expert routing
observed WSI tokens
Residual compensation
gate $\sigma(\mathbf{w}_W{}^\top \mathbf{c} + b_W)$
bank $\mathbf{B}_W$ (learned)
mean
$\mathbf{W}_W \mathbf{c}$
compensation $\hat{\mathbf{u}}_W$
$\mathbf{e}_{\text{obs}}$
$\mathbf{e}_{\text{miss}}$
$\boldsymbol{\pi}(\mathbf{x})$
router
4 experts
$a_W = 1$
$a_W = 0$
$\mathscr{L}_{\text{distil}}$
Cross-modal fusion
availability-aware expert routing
residual compensation
main path
missing modality only
training only
availability
observed routing
compensation routing
compensation tokens
frozen
sum
product